

System Identification Method for Robotic Manipulator based on Dynamic Momentum Regressor

Kyongho Park and Youngjin Choi

Abstract—The paper presents a momentum regressor using both numerical integration and Hamiltonian dynamic description for robotic manipulator. It has an advantage in that the proposed momentum regressor does not require the acceleration estimation or sensor for the identification of dynamic parameters. In addition, the identification algorithm is proposed by using Lagrange multiplier and bordered gramian matrix for the minimization of difference between the previous and the current estimates of dynamic parameters. The developed algorithm is intuitive and easy to implement in real-time fashion. Finally, the effectiveness of the proposed momentum regressor and identification method is shown through the numerical simulations about a planar two-link robotic manipulator system.

I. INTRODUCTION

For precise control of robotic manipulators, exact knowledge about robotic motion description including its dynamic parameters such as link inertias, masses and friction coefficients are required. For this purpose, an appropriate identification algorithm is also needed because various friction coefficients are mainly varying according to the assembling and manufacturing procedures, even though the dynamic parameters are considered from the initial design phase. A variety of identification schemes for linear and nonlinear systems have suggested in [1]. In addition, since the identification on the dynamic parameters of robotic manipulator has been one of the classical research topics, several papers and textbooks have dealt with the topic [2], [3], [4], [5]. In the paper, novel identification method is proposed by using new form of regressor, Lagrange multiplier and bordered gramian matrix through the minimization of difference between the previous and the current estimations of dynamic parameters. In the previous researches, the actual acceleration information was replaced into the desired acceleration one thanks to the adaptive control scheme. However, the paper does not contain the desired trajectory and the adaptive control scheme, but focuses only on the identification itself.

Since the identification of dynamic parameters for robotic manipulator is basically based on the classical mechanics (equations of motion), let us review well-known three kinds of dynamic descriptions such as Newton-Euler equation, Euler-Lagrange equation, and the principle of virtual work.

K. Park is with the School of Mechanical Engineering (also he belongs to the LS Mtron Co.), POSTECH, 37673, South Korea. e-mail: neotele@lsmtron.com

Y. Choi is with the School of Electrical Engineering, Hanyang University, 15588, South Korea. e-mail: cyj@hanyang.ac.kr

This work was supported in part by the Convergence Technology Development Program for Bionic Arm through the National Research Foundation of Korea funded by the Ministry of Science, ICT & Future Planning (NRF-2015M3C1B2052811), and in part by the National Research Foundation of Korea (NRF-2013R1A1A2010192), Republic of Korea.

Then the second-order differential equations are obtained and the algorithms come down to calculating and integrating the acceleration [6]. Actually, aforementioned three methods result in the minimal representation for the corresponding equation of motion if there are not non-holonomic constraints. Generally, expressing the equations of motion with a minimal set of coordinates results in less differential equations, which are, however, more coupled and usually exhibit stronger nonlinearities compared to the non-minimal formulations [7]. The representative non-minimal formulation is the Hamiltonian equation of motion. Its advantage is that it brings a set of first-order differential equations, though it results in the mixed form of differential equations. Actually the Hamiltonian equation of motion is a dual form of the Euler-Lagrange ones because the Legendre transform could be always found due to the positiveness of inertia matrix. In order to define a momentum regressor as new form, Hamiltonian equation of motion will be utilized for further mathematical derivations.

The simulation and control of robotic manipulators demand the knowledge on the values of dynamic parameters. Computing such parameters from the design data of the mechanical structure is not simple, as an alternative, we may adopt CAD (computer aided design) tools which allow computing the values of dynamic parameters (links, actuators, and transmissions) on the basis of their geometry and the type of material employed. Nevertheless, the estimates obtained by using such techniques are inaccurate because of the simplification typically introduced by geometric modeling. Moreover, the complex dynamic effects, such as a variety of joint frictions, may not be taken into account. Thus the robust identification algorithm for the dynamic parameters is required. In order to identify the dynamic parameters of robot manipulator, first, we establish the regressor from the corresponding equation of motion. In [2], the acceleration-based regressor was derived from the conventional Newton-Euler algorithm. In this paper, the momentum regressor is newly defined by using the Hamiltonian equation motion, and it does not require the acceleration measurement or estimation for robotic manipulator.

This paper is organized as follows: section II derives momentum regressor from the classical robot dynamics, section III suggests the real-time identification method by using the optimization subject to the equality constraint, the numerical simulations are shown in section IV, and then we draw the conclusions in section V.

基于动态动量回归器的机器人机械臂系统辨识方法

朴景浩 和 崔永镇

Abstract—— 论文提出了一种使用数值积分和哈密顿动力学描述的机器人操作器动量回归器。其优点在于所提出的动量回归器在识别动力学参数时不需要加速度估计或传感器。此外，提出了一种利用拉格朗日乘子和带边界格拉姆矩阵的识别算法，以最小化先前与当前动力学参数估计值之间的差异。所开发的算法直观且易于实时实现。最后，通过关于二维平面两连杆机器人操作器系统的数值仿真，展示了所提出的动量回归器和识别方法的有效性。

一、引言

为了精确控制机器人操作器，需要对机器人运动描述有精确的了解，包括其动态参数，如连杆惯性、质量和摩擦系数。为此，还需要合适的识别算法，因为各种摩擦系数主要因装配和制造工艺的不同而变化，即使动态参数从初始设计阶段就已考虑。在文献[1]中提出了各种用于线性和非线性系统的识别方案。此外，由于机器人操作器动态参数的识别一直是经典研究课题，已有多篇论文和教科书涉及该主题[2]、[3]、[4]、[5]。在本文中，提出了一种新型识别方法，通过最小化先前与当前动态参数估计之间的差异，利用新的回归形式、拉格朗日乘子和边界Gramian矩阵进行识别。

由于机器人机械臂的动态参数识别基本上是基于经典力学（运动方程），让我们回顾三种著名的动态描述方法，如牛顿-欧拉方程、欧拉-拉格朗日方程以及虚功原理。

K. Park is with the School of Mechanical Engineering (also he belongs to the LS Mtron Co.), POSTECH, 37673, South Korea. e-mail: neotele@lsmtron.com

Y. Choi is with the School of Electrical Engineering, Hanyang University, 15588, South Korea. e-mail: cyj@hanyang.ac.kr

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然后得到二阶微分方程，算法归结为计算和积分加速度 [6]。实际上，如果不存在非完整约束，上述三种方法会得到相应运动方程的最小表示。一般来说，用最小坐标集表示运动方程会产生更少的微分方程，但这些方程相互耦合更紧密，并且相比非最小形式通常表现出更强的非线性 [7]。具有代表性的非最小形式是哈密顿运动方程。它的优点是生成了一组一阶微分方程，尽管它导致微分方程的混合形式。实际上，哈密顿运动方程是欧拉-拉格朗日方程的对偶形式，因为由于惯性矩阵的正定性，总是可以找到勒让德变换。为了定义新的动量回归矩阵

机器人操作器的仿真与控制需要了解动态参数的数值。从机械结构的设计数据计算这些参数并不简单，作为一种替代方法，我们可以采用 CAD（计算机辅助设计）工具，根据几何形状和所用材料类型计算动态参数（连杆、执行器和传动装置）的数值。然而，由于几何建模通常引入的简化，通过这种技术获得的估算是不准确的。此外，复杂的动态效应，如各种关节摩擦，可能无法被考虑。因此，需要一个用于动态参数的鲁棒识别算法。为了识别机器人操作器的动态参数，首先，我们从相应的运动方程建立回归器。在[2]中，加速度基回归器是从传统的牛顿-欧拉方法推导出来的。

本文组织如下：第二节从经典机器人动力学推导动量回归器，第三节通过使用受等式约束的优化提出实时识别方法，第四节展示数值仿真，然后在第五节得出结论。

II. MOMENTUM REGRESSOR

The robotic manipulator is generally described using Euler-Lagrange equation of motion as following form:

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + \tau_f(q, \dot{q}) + J^T(q)f_{ext} \quad (1)$$

where $q = [q_1, q_2, \dots, q_n]^T$ is an n -dimensional configuration (or joint displacement) vector, $\dot{q}$ a joint velocity vector, $\ddot{q}$ a joint acceleration vector, $M(q)$ an inertia matrix, $C(q, \dot{q})$ a Coriolis and centrifugal matrix, $g(q)$ a gravity vector, $\tau_f(q, \dot{q})$ a frictional torque vector, $J(q)$ a manipulator Jacobian, f_{ext} an external force such as load, and τ a control input torque vector. The natural choice of a model for identification of dynamic parameters is above Euler-Lagrange model of the manipulator, which has the property of linearity in dynamic parameters as following form:

$$\tau = Y_L(q, \dot{q}, \ddot{q})\theta, \quad (2)$$

where $Y_L(q, \dot{q}, \ddot{q})$ is a conventional regressor matrix containing all the nonlinear terms related to the displacement, velocity and acceleration, and θ is a constant dynamic parameter vector such as link inertia, link mass and positions of center of mass, and friction coefficients. The general method constructing the conventional regressor has been suggested in [2], [3]. It is referred to as a Lagrange regressor or conventional regressor. If the control input torque, displacement, velocity, acceleration vectors are all measurable every time instants, then it might be possible to identify the dynamic parameters θ of robotic manipulator, but the acceleration information should be provided for the algorithm implementation.

Since the robotic manipulator system is not normally equipped with the accelerometer, we need to construct the estimator to make acceleration signal from the velocity or displacement through numerical differentiation. It might bring wrong identification results. On the other hand, even though it has an accelerometer, it is not easy to combine different information obtained from two or more sensors because the measurement bandwidths of the sensors are different each other. Thus many previous simulations and experiments might bring inexact identification results. As an alternative, new form of regressor that does not require the numerical differentiation or acceleration information is proposed in the following.

Let us define the momentum vector as follow:

$$p = M(q)\dot{q}, \quad (3)$$

where p is an n -dimensional momentum vector derived from the product of inertia matrix and joint velocity vector. Also the above is referred to as a Legendre transformation and it is always possible if and only if the inertia matrix is positive definite and symmetric. In addition, we know that the time derivative of the inertia matrix has the following relation with Coriolis and centrifugal matrix [3]:

$$\dot{M}(q) = C(q, \dot{q}) + C^T(q, \dot{q}), \quad (4)$$

where it can be interpreted that the time derivative of inertia matrix is equal to two times of even part of Coriolis and centrifugal matrix.

After taking a time derivative of Eq. (3), if we rearrange it by using Eqs. (1) and (4), then we have

$$\begin{aligned} \dot{p} &= \dot{M}(q)\dot{q} + M\ddot{q} \\ &= [C(q, \dot{q}) + C^T(q, \dot{q})]\dot{q} \\ &\quad - [C(q, \dot{q})\dot{q} + g(q) + \tau_f(q, \dot{q}) + J^T(q)f_{ext} - \tau] \\ &= C^T(q, \dot{q})\dot{q} - g(q) - \tau_f(q, \dot{q}) - J^T(q)f_{ext} + \tau, \end{aligned} \quad (5)$$

where the above implies the Hamiltonian equation of motion that does not explicitly include an acceleration term. This is the reason why we utilize the Hamiltonian representation for equation of motion. Now let us define first subregressor as

$$\dot{p} = W_1(q, \dot{q})\theta + \tau, \quad (6)$$

where

$$W_1(q, \dot{q})\theta \triangleq C^T(q, \dot{q})\dot{q} - g(q) - \tau_f(q, \dot{q}) - J^T(q)f_{ext}, \quad (7)$$

in which note that the first subregressor is not dependent on the acceleration. Also second subregressor can be defined from the momentum equation as:

$$\begin{aligned} p &= M(q)\dot{q} \\ &\triangleq W_2(q, \dot{q})\theta. \end{aligned} \quad (8)$$

After integrating both sides of Eq. (6), if Eq. (8) is applied, then the new form of regressor is obtained as:

$$u(t) = Y_H(q, \dot{q}, t)\theta, \quad (9)$$

where

$$u(t) \triangleq \int_0^t \tau \, dt \quad (10)$$

$$Y_H(q, \dot{q}, t) \triangleq W_2(q, \dot{q}) - \int_0^t W_1(q, \dot{q}) \, dt \quad (11)$$

in which $u(t)$ implies the time integration of control input torque from the initial time 0 to the current time t , and $Y_H(q, \dot{q}, t)$ is referred to as “momentum regressor” composed of three independent variables such as q , $\dot{q}$, and time t in the paper. If above momentum regressor is utilized, then we know that it does not require the acceleration information but the numerical integration of the applied input torques and first subregressor. The numerical integration is more stable and exact than the numerical differentiation in viewpoint of noise removal. As a matter of fact, Eq. (9) is equal to the time integration form of conventional Lagrange regressor of Eq. (2).

The following section deals with an identification method obtained by applying the Lagrange multiplier and bordered gramian methods in order to solve the optimization problem.

II. 动量回归器

机械手通常使用欧拉-拉格朗日运动方程描述，形式如下：

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + \tau_f(q, \dot{q}) + J^T(q)f_{ext} \quad (1)$$

其中 $q = [q_1, q_2, \dots, q_n]^T$ 是一个 n 维的配置（或关节位移）向量， $\dot{q}$ 是关节速度向量， $\ddot{q}$ 是关节加速度向量， $M(q)$ 是惯性矩阵， $C(q, \dot{q})$ 是科里奥利和离心矩阵， $g(q)$ 是重力向量， $\tau_f(q, \dot{q})$ 是摩擦力矩向量， $J(q)$ 是机械臂雅可比矩阵， f_{ext} 是外力如负载， τ 是控制输入力矩向量。用于动态参数识别的模型的自然选择是上述机械臂的欧拉-拉格朗日模型，该模型具有如下形式的动态参数线性特性：

$$\tau = Y_L(q, \dot{q}, \ddot{q})\theta, \quad (2)$$

其中 $Y_L(q, \dot{q}, \ddot{q})$ 是包含与位移、速度和加速度相关的所有非线性项的常规回归矩阵，而 θ 是常数动力学参数向量，例如连杆惯量、连杆质量和质心位置，以及摩擦系数。[2]、[3] 中提出了构建常规回归矩阵的一般方法。它被称为拉格朗日回归矩阵或常规回归矩阵。如果控制输入力矩、位移、速度、加速度向量在每个时间点都可以测量，那么可能有办法识别机械手的动力学参数 θ ，但算法实现应提供加速度信息。

由于机器人操作系统通常没有配备加速度计，我们需要构建估计器，通过数值微分从速度或位移中得到加速度信号。这可能会带来错误的识别结果。另一方面，即使有加速度计，由于传感器的测量带宽彼此不同，将来自两个或更多传感器的不同信息结合起来也并不容易。因此，许多之前的仿真和实验可能会产生不准确的识别结果。作为替代，以下提出了一种不需要数值微分或加速度信息的新形式回归器。

让我们将动量矢量定义如下：

$$p = M(q)\dot{q}, \quad (3)$$

其中 p 是一个 n 维的动量向量，由惯性矩阵与关节速度向量的乘积得到。此外，上述内容被称为勒让德变换，并且当且仅当惯性矩阵是正定且对称的时，这种变换才总是可能的。此外，我们知道惯性矩阵的时间导数与科里奥利矩阵和离心矩阵具有以下关系 [3]：

$$\dot{M}(q) = C(q, \dot{q}) + C^T(q, \dot{q}), \quad (4)$$

可以解释为惯性矩阵的时间导数等于科里奥利力和离心力矩阵偶数部分的两倍。

对方程 (3) 进行时间求导后，如果我们利用方程 (1) 和 (4) 对其重新排列，那么我们得到

$$\begin{aligned} \dot{p} &= \dot{M}(q)\dot{q} + M\ddot{q} \\ &= [C(q, \dot{q}) + C^T(q, \dot{q})]\dot{q} \\ &\quad - [C(q, \dot{q})\dot{q} + g(q) + \tau_f(q, \dot{q}) + J^T(q)f_{ext} - \tau] \\ &= C^T(q, \dot{q})\dot{q} - g(q) - \tau_f(q, \dot{q}) - J^T(q)f_{ext} + \tau, \end{aligned} \quad (5)$$

上述表达式意味着不显式包含加速度项的哈密顿运动方程。这就是我们为何使用哈密顿表示法来描述运动方程的原因。现在我们首先定义第一个子回归器为

$$\dot{p} = W_1(q, \dot{q})\theta + \tau, \quad (6)$$

哪里

$$W_1(q, \dot{q})\theta \triangleq C^T(q, \dot{q})\dot{q} - g(q) - \tau_f(q, \dot{q}) - J^T(q)f_{ext}, \quad (7)$$

需要注意的是，第一个子回归量不依赖于加速度。同时，第二个子回归量可以从动量方程中定义为：

$$\begin{aligned} p &= M(q)\dot{q} \\ &\triangleq W_2(q, \dot{q})\theta. \end{aligned} \quad (8)$$

在对方程 (6) 的两边进行积分后，如果应用方程 (8)，则得到回归量的新形式为：

$$u(t) = Y_H(q, \dot{q}, t)\theta, \quad (9)$$

哪里

$$u(t) \triangleq \int_0^t \tau \, dt \quad (10)$$

$$Y_H(q, \dot{q}, t) \triangleq W_2(q, \dot{q}) - \int_0^t W_1(q, \dot{q}) \, dt \quad (11)$$

其中 $u(t)$ 表示从初始时间 0 到当前时间 t 的控制输入扭矩的时间积分， $Y_H(q, \dot{q}, t)$ 被称为“动量回归器”，由文中三个独立变量组成，如 q 、 $\dot{q}$ 和时间 t 。如果使用上述动量回归器，那么我们知道它不需要加速度信息，而需要的是施加的输入扭矩和第一个子回归器的数值积分。从噪声消除的角度来看，数值积分比数值微分更稳定且更精确。实际上，公式 (9) 等价于公式 (2) 的传统拉格朗日回归器的时间积分形式。

以下部分介绍了一种识别方法，该方法通过应用拉格朗日乘子法和边界格拉姆方法来解决优化问题。

III. IDENTIFICATION ALGORITHM USING MOMENTUM REGRESSOR

For given joint displacement and velocity, the momentum regressor can be determined as a real-time form. Furthermore, the pairs of the integration of input torque and the momentum regressor, $\{u[i], Y_H[i]\}$ at the i th sample, are sequentially obtained as follows:

$$u[i] = Y_H[i]\theta, \quad (12)$$

where

$$Y_H[i] = W_2[i] - \sum_{k=1}^i W_1[k]\delta t \quad (13)$$

$$u[i] = \sum_{k=1}^i \tau[k]\delta t \quad (14)$$

in which δt implies the sampling time interval, namely, time $t = i \cdot \delta t$. Since the measurements $\{u[i], Y_H[i]\}$ for $i = 1, 2, \dots, n$ contain the information about constant dynamic parameters to be identified, it is natural to choose new estimate vector $\hat{\theta}[i]$ so as to minimize the parameter variation energy satisfying the constraint (12). This is a typical least squares problem in [8]. First let us make a normal equation by multiplying $Y_H^T[i]$ with both sides of Eq. (12) as follow:

$$Y_H^T[i]u[i] = (Y_H^T[i]Y_H[i])\hat{\theta}[i] \quad \text{for } i = 1, 2, \dots \quad (15)$$

Second, since above equality should be satisfied for all time instants, it can be accumulated till the current sample i as follow:

$$\sum_{k=1}^i Y_H^T[k]u[k] = \left(\sum_{k=1}^i Y_H^T[k]Y_H[k] \right) \hat{\theta}[i]. \quad (16)$$

On the other hand, since the parameter vector $\hat{\theta}[i]$ to be identified should arrive at the constant values, it can be formulated as the minimization of parameter variation energy with the equality constraint of Eq. (16) as follows:

$$\begin{aligned} \min \frac{1}{2} \left\| \hat{\theta}[i] - \hat{\theta}[i-1] \right\|^2 \\ \text{subject to } \Gamma[i] = \Omega[i]\hat{\theta}[i] \end{aligned} \quad (17)$$

where

$$\Gamma[i] \triangleq \sum_{k=1}^i Y_H^T[k]u[k] \quad (18)$$

$$\Omega[i] \triangleq \sum_{k=1}^i Y_H^T[k]Y_H[k]. \quad (19)$$

Now if we introduce a Lagrangian multiplier method to handle above minimization, we have

$$\begin{aligned} L = \frac{1}{2} \left(\hat{\theta}[i] - \hat{\theta}[i-1] \right)^T \left(\hat{\theta}[i] - \hat{\theta}[i-1] \right) \\ + \lambda^T \left(\Gamma[i] - \Omega[i]\hat{\theta}[i] \right), \end{aligned} \quad (20)$$

where λ is a Lagrange multiplier vector with appropriate dimension. Taking the derivatives with respect to $\hat{\theta}[i]$ and λ , we have

$$\hat{\theta}[i] - \hat{\theta}[i-1] - \Omega^T[i]\lambda = 0 \quad (21)$$

$$\Gamma[i] - \Omega[i]\hat{\theta}[i] = 0. \quad (22)$$

By rearranging above two equations, we can find the matrix-vector equation

$$\begin{bmatrix} I & \Omega^T[i] \\ \Omega[i] & 0 \end{bmatrix} \begin{bmatrix} \hat{\theta}[i] \\ -\lambda \end{bmatrix} = \begin{bmatrix} \hat{\theta}[i-1] \\ \Gamma[i] \end{bmatrix}, \quad (23)$$

where the first matrix is called as ‘‘bordered gramian matrix’’ in [9], [10]. Exact solution of Eq. (23) is given by inverting the bordered gramian matrix. In addition, the concrete procedure to obtain the inverse of the bordered gramian matrix was suggested in [10] and its inverse has the following form:

$$\begin{bmatrix} I & \Omega^T[i] \\ \Omega[i] & 0 \end{bmatrix}^{-1} = \begin{bmatrix} W[i] - W[i]\Omega[i]^T Z[i]\Omega[i]W[i] & W[i]\Omega[i]^T Z[i] \\ Z[i]\Omega[i]^T W[i] & I - Z[i] \end{bmatrix} \quad (24)$$

where

$$W[i] \triangleq (I + \Omega[i]^T \Omega[i])^{-1} \quad (25)$$

$$Z[i] \triangleq (\Omega[i]W[i]\Omega[i]^T)^+, \quad (26)$$

in which the existence of the matrix $W[i]$ is always guaranteed but the matrix $Z[i]$ exists only when $\Omega[i]$ has full rank. Thus we have defined the matrix $Z[i]$ by using the pseudoinverse obtained by using the singular value decomposition, not the inverse. Indeed even though the full rank $Y_H[i]$ is guaranteed, we cannot guarantee full rank $Z[i]$ because the normal equation is an expansion of $Y_H[i]$ towards higher dimension of rectangular $Y_H[i]$. Strictly speaking, the full rank condition of $Z[i]$ is not equal to the persistent excitation (PE) condition in the adaptive control [11], [12], because the PE condition implies just the full rank of $Y_H[i]$ but $Z[i]$ makes use of the summation of normal form of $Y_H[i]$.

Now let us obtain the solution of Eq. (23)

$$\hat{\theta}[i] = \Omega_W^+[i]\Gamma[i] + (I - \Omega_W^+[i]\Omega[i])W[i]\hat{\theta}[i-1] \quad (27)$$

$$\lambda = (I - Z[i])\Gamma[i] + Z[i]\Omega[i]^T W[i]\hat{\theta}[i-1]$$

where

$$\begin{aligned} \Omega_W^+[i] &\triangleq W[i]\Omega[i]^T Z[i] \\ &= W[i]\Omega[i]^T (\Omega[i]W[i]\Omega[i]^T)^+. \end{aligned} \quad (28)$$

Fig. 1 illustrates the proposed identification scheme of Eq. (27). Since the matrix $W[i]$ has the full rank, the following statements are always satisfied in Fig. 1.

- 1) $\mathcal{R}(\Omega_W^+[i])$ and $\mathcal{N}(\Omega[i])$ are $W^{-1}[i]$ weighted orthogonal because

$$\begin{aligned} &(\Omega_W^+[i])^T W^{-1}[i](I - \Omega^+[i]\Omega[i]) \\ &= (\Omega[i]W[i]\Omega[i]^T)^+ \Omega[i]W[i]W^{-1}[i](I - \Omega^+[i]\Omega[i]) \\ &= (\Omega[i]W[i]\Omega[i]^T)^+ \Omega[i](I - \Omega^+[i]\Omega[i]) \\ &= 0. \end{aligned}$$

III. 使用动量回归器的识别算法

对于给定的关节位移和速度，动量回归量可以确定为实时形式。此外，输入力矩与动量回归量积分的对，即在第 i 个样本处的 $\{u[i], Y_H[i]\}$ ，按顺序获得如下：

$$u[i] = Y_H[i]\theta, \quad (12)$$

哪里

$$Y_H[i] = W_2[i] - \sum_{k=1}^i W_1[k]\delta t \quad (13)$$

$$u[i] = \sum_{k=1}^i \tau[k]\delta t \quad (14)$$

其中 δt 表示采样时间间隔，即时间 $t = i \cdot \delta t$ 。由于测量 $\{u[i], Y_H[i]\}$ 对于 $i = 1, 2, \dots, n$ 包含了待识别的恒定动态参数的信息，因此自然选择新的估计向量 $\hat{\theta}[i]$ 以最小化满足约束 (12) 的参数变化能量。这是 [8] 中的典型最小二乘问题。首先，通过将 $Y_H^T[i]$ 与方程 (12) 的两边相乘，我们可以得到一个法方程，如下所示：

$$Y_H^T[i]u[i] = (Y_H^T[i]Y_H[i])\hat{\theta}[i] \quad \text{for } i = 1, 2, \dots \quad (15)$$

其次，由于上述等式应在所有时间点上成立，因此可以累积到当前样本 i ，如下所示：

$$\sum_{k=1}^i Y_H^T[k]u[k] = \left(\sum_{k=1}^i Y_H^T[k]Y_H[k] \right) \hat{\theta}[i]. \quad (16)$$

另一方面，由于待识别的参数向量 $\hat{\theta}[i]$ 应达到常数值，因此可以将其表述为在等式约束 Eq. (16) 下的参数变化能量最小化，如下所示：

$$\begin{aligned} \min \frac{1}{2} \left\| \hat{\theta}[i] - \hat{\theta}[i-1] \right\|^2 \\ \text{subject to } \Gamma[i] = \Omega[i]\hat{\theta}[i] \end{aligned} \quad (17)$$

哪里

$$\Gamma[i] \triangleq \sum_{k=1}^i Y_H^T[k]u[k] \quad (18)$$

$$\Omega[i] \triangleq \sum_{k=1}^i Y_H^T[k]Y_H[k]. \quad (19)$$

现在，如果我们引入拉格朗日乘子法来处理上述最小化问题，我们得到

$$\begin{aligned} L = \frac{1}{2} \left(\hat{\theta}[i] - \hat{\theta}[i-1] \right)^T \left(\hat{\theta}[i] - \hat{\theta}[i-1] \right) \\ + \lambda^T \left(\Gamma[i] - \Omega[i]\hat{\theta}[i] \right), \end{aligned} \quad (20)$$

其中 λ 是具有适当维度的拉格朗日乘子向量。对 $\hat{\theta}[i]$ 和 λ 求导数，我们得到

$$\hat{\theta}[i] - \hat{\theta}[i-1] - \Omega^T[i]\lambda = 0 \quad (21)$$

$$\Gamma[i] - \Omega[i]\hat{\theta}[i] = 0. \quad (22)$$

通过重新排列上述两个方程，我们可以得到矩阵-向量方程

$$\begin{bmatrix} I & \Omega^T[i] \\ \Omega[i] & 0 \end{bmatrix} \begin{bmatrix} \hat{\theta}[i] \\ -\lambda \end{bmatrix} = \begin{bmatrix} \hat{\theta}[i-1] \\ \Gamma[i] \end{bmatrix}, \quad (23)$$

第一个矩阵在[9], [10]中称为“边界格拉姆矩阵”。方程(23)的精确解是通过边界格拉姆矩阵求逆给出的。此外，在[10]中提出了获得边界格拉姆矩阵逆矩阵的具体步骤，其逆矩阵具有以下形式：

$$\begin{bmatrix} I & \Omega^T[i] \\ \Omega[i] & 0 \end{bmatrix}^{-1} = \begin{bmatrix} W[i] - W[i]\Omega[i]^T Z[i]\Omega[i]W[i] & W[i]\Omega[i]^T Z[i] \\ Z[i]\Omega[i]^T W[i] & I - Z[i] \end{bmatrix} \quad (24)$$

哪里

$$W[i] \triangleq (I + \Omega[i]^T \Omega[i])^{-1} \quad (25)$$

$$Z[i] \triangleq (\Omega[i]W[i]\Omega[i]^T)^+, \quad (26)$$

其中矩阵 $W[i]$ 的存在总是有保证的，但矩阵 $Z[i]$ 仅在 $\Omega[i]$ 满秩时存在。因此，我们通过使用奇异值分解得到的伪逆而不是逆矩阵来定义矩阵 $Z[i]$ 。实际上，即使保证了满秩的 $Y_H[i]$ ，我们也不能保证 $Z[i]$ 的满秩性，因为正规方程是 $Y_H[i]$ 向高维矩形 $Y_{Hl}[i]$ 的扩展。严格来说， $Z[i]$ 的满秩条件并不等同于自适应控制中的持续激励 (PE) 条件 [11], [12]，因为 PE 条件仅意味着 $Y_H[i]$ 的满秩性，而 $Z[i]$ 利用了 $Y_{Hl}[i]$ 的正规形式求和。

现在让我们求解方程 (23) 的解

$$\begin{aligned} \hat{\theta}[i] &= \Omega_W^+[i]\Gamma[i] + (I - \Omega_W^+[i]\Omega[i])W[i]\hat{\theta}[i-1] \\ \lambda &= (I - Z[i])\Gamma[i] + Z[i]\Omega[i]^T W[i]\hat{\theta}[i-1] \end{aligned} \quad (27)$$

哪里

$$\begin{aligned} \Omega_W^+[i] &\triangleq W[i]\Omega[i]^T Z[i] \\ &= W[i]\Omega[i]^T (\Omega[i]W[i]\Omega[i]^T)^+. \end{aligned} \quad (28)$$

图 1 说明了方程 (27) 提出的识别方案。由于矩阵 $W[i]$ 是满秩的，因此图 1 中总是满足以下条件。

1) $\mathcal{R}(\Omega_W^+[i])$ 和 $\mathcal{R}(\Omega[i])$ 是 $W^{-1}[i]$ 加权正交的，因为

$$\begin{aligned} &(\Omega_W^+[i])^T W^{-1}[i](I - \Omega^+[i]\Omega[i]) \\ &= (\Omega[i]W[i]\Omega[i]^T)^+ \Omega[i]W[i]W^{-1}[i](I - \Omega^+[i]\Omega[i]) \\ &= (\Omega[i]W[i]\Omega[i]^T)^+ \Omega[i](I - \Omega^+[i]\Omega[i]) \\ &= 0. \end{aligned}$$

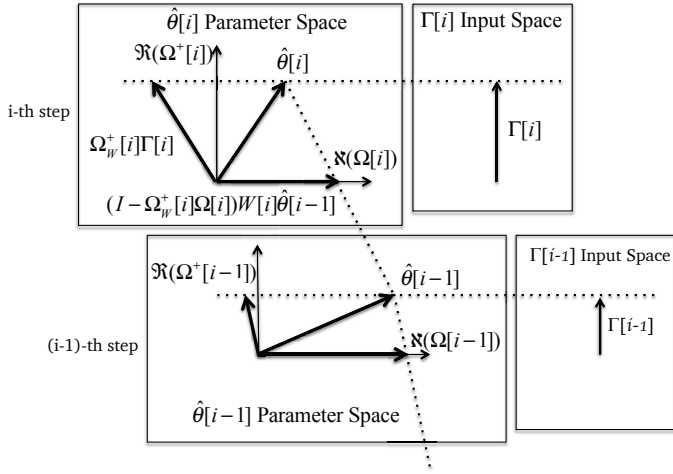


Fig. 1: Geometric interpretation of Eq. (27)

- 2) $\mathcal{R}(\Omega^T[i])$ and $\mathcal{N}(\Omega[i])$ are geometrically orthogonal because

$$(\Omega^T[i])^T (I - \Omega^+[i]\Omega[i]) = \Omega[i](I - \Omega^+[i]\Omega[i]) = 0.$$

We should note that $\mathcal{R}(\Omega^+[i]) = \mathcal{R}(\Omega^T[i])$, $\mathcal{R}(\cdot)$ and $\mathcal{N}(\cdot)$ mean the range space and the null space, respectively. An item 1) implies the orthogonality in the weighted metric and another item 2) implies the geometrical orthogonality between the range space of $\Omega^T[i]$ and null space of $\Omega[i]$.

Since the geometrical orthogonality between $\mathcal{R}(\Omega^+[i])$ and $\mathcal{N}(\Omega[i])$ is satisfied in the parameter space $\hat{\theta}[i]$, we have expressed them as coordinate axes in the orthogonal coordinate system as shown in Fig. 1. The identification algorithm of Eq. (27) is composed of the sum of the component $\Omega_W^+[i]\Gamma[i]$ embedded on $\mathcal{R}(\Omega_W^+[i])$ and another component $(I - \Omega_W^+[i]\Omega[i])W[i]\hat{\theta}[i-1]$ embedded on $\mathcal{N}(\Omega[i])$ which is not geometrically orthogonal to $\mathcal{R}(\Omega_W^+[i])$ but $W^{-1}[i]$ weighted orthogonal to $\mathcal{R}(\Omega_W^+[i])$. The matrix $W^{-1}[i]$ becomes able to rotate the embedded range space with suitable rotation angle, in other words, it changes the orthogonal null space projection into the inclined null space projection efficient for utilization of previous parameter estimate $\hat{\theta}[i-1]$. On the other hand, if $\Omega[i]$ has the full rank, the null space does not exist and the inclined null space projection disappears in the identification algorithm of Eq. (27).

As summary, the Eq. (27) aims at obtaining the optimal estimates of the dynamic parameters by using the mapping $\Omega[i]$ from the input $\Gamma[i]$. During the identification process, if the mapping $\Omega[i]$ loses the rank, then the previous parameter estimates are utilized through the inclined null space projection scheme, ultimately, for improving both identification accuracy and convergence speed. The next section shows the effectiveness of the proposed identification algorithm through several simulations.

IV. SIMULATIONS

Let us assume that a planar two-link manipulator is handling the unknown mass M at the end-effector as shown

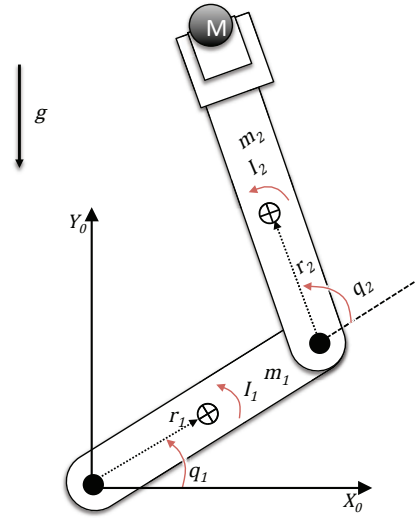


Fig. 2: Typical planar two-link manipulator handling unknown mass M

in Fig. 2, in order for verification of the proposed identification algorithm. The dynamics of Eq. (1) has the following components:

$$M(q) = \begin{bmatrix} I_1 + I_2 + m_2 l_1^2 + 2m_2 r_2 l_1 c_2 & I_2 + m_2 r_2 l_1 c_2 \\ I_2 + m_2 r_2 l_1 c_2 & I_2 \end{bmatrix} \quad (29)$$

$$C(q, \dot{q}) = \begin{bmatrix} -m_2 r_2 l_1 s_2 \dot{q}_2 & -m_2 r_2 l_1 s_2 (\dot{q}_1 + \dot{q}_2) \\ m_2 r_2 l_1 s_2 \dot{q}_1 & 0 \end{bmatrix} \quad (30)$$

$$g(q) = \begin{bmatrix} -m_1 r_1 g c_1 - m_2 l_1 g c_1 - m_2 r_2 g c_{12} \\ -m_2 r_2 g c_{12} \end{bmatrix} \quad (31)$$

$$\tau_f(q, \dot{q}) = \begin{bmatrix} F_{s1} \text{sign}(\dot{q}_1) + F_{c1} \dot{q}_1 \\ F_{s2} \text{sign}(\dot{q}_2) + F_{c2} \dot{q}_2 \end{bmatrix} \quad (32)$$

$$J(q) = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix} \quad (33)$$

$$f_{ext} = \begin{bmatrix} 0 \\ -Mg \end{bmatrix} \quad (34)$$

where I_1 and I_2 are the inertias of link 1 and 2, respectively, m_1 and m_2 are the masses of link 1 and 2, respectively, l_1 and l_2 are the lengths of link 1 and 2, respectively, r_1 and r_2 are the positions of center of masses of link 1 and 2, respectively, q_1 and q_2 are the joint angles of link 1 and 2, respectively, F_{s1} and F_{s2} are the stiction coefficients of link 1 and 2, respectively, F_{c1} and F_{c2} are the coulomb friction coefficients of link 1 and 2, respectively, g implies the gravitational acceleration constant, $g = 9.8[m/s^2]$, and

$$\begin{aligned} c_1 &= \cos(q_1) & c_2 &= \cos(q_2) \\ c_{12} &= \cos(q_1 + q_2) & s_1 &= \sin(q_1) \\ s_2 &= \sin(q_2) & s_{12} &= \sin(q_1 + q_2). \end{aligned}$$

For the simulation study, the dynamic parameters are set as $l_1 = l_2 = 0.5[m]$, $m_1 = m_2 = 1.0[kg]$, $r_1 = r_2 = 0.2[m]$, $I_1 = I_2 = 0.05[kg \cdot m^2]$, $F_{s1} = F_{s2} = 0.1$, $F_{c1} = F_{c2} = 0.1$, and $M = 0.1[kg]$. Also these parameters are treated as the unknowns to be identified later.

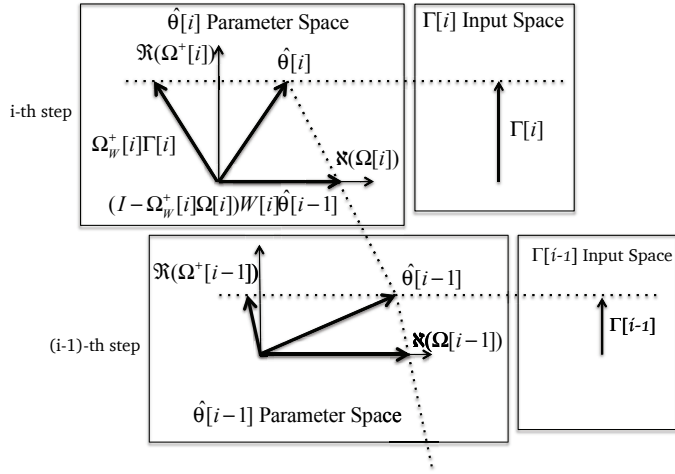


图1: 方程(27)的几何解释

2) $\mathcal{R}(\Omega^T[i])$ 和 $\mathcal{N}(\Omega[i])$ 在几何上是正交的, 因为

$$(\Omega^T[i])^T(I - \Omega^+[i]\Omega[i]) = \Omega[i](I - \Omega^+[i]\Omega[i]) = 0.$$

我们应该注意, $\mathcal{R}(\Omega^+[i]) = \mathcal{R}(\Omega^T[i])$, $\mathcal{R}(\cdot)$ 和 $\mathcal{N}(\cdot)$ 分别表示值域空间和零空间。项 1) 表示加权度量下的正交性, 而另一项 2) 表示 $\Omega^T[i]$ 的值域空间与 $\Omega[i]$ 的零空间之间的几何正交性。

由于在参数空间 $\hat{\theta}[i]$ 中满足 $\mathcal{R}(\Omega^+[i])$ 与 $\mathcal{N}(\Omega[i])$ 之间的几何正交性, 我们已将它们表示为如图 1 所示的正交坐标系中的坐标轴。方程 (27) 的识别算法由嵌入在 $\mathcal{R}(\Omega_w^+[i])$ 上的分量 $\Omega_w^+[i]\Gamma[i]$ 与另一个嵌入在 $\mathcal{N}(\Omega[i])$ 上的分量 $(I - \Omega_w^+[i]\Omega[i])W[i]\hat{\theta}[i-1]$ 的和组成, 该分量在几何上不与 $\mathcal{R}(\Omega_w^+[i])$ 正交, 但在 $\mathcal{R}(\Omega_w^+[i])$ 上是 $W^{-1}[i]$ 加权正交的。矩阵 $W^{-1}[i]$ 能够以适当的旋转角旋转嵌入的范围空间, 换句话说, 它将正交零空间投影改为倾斜零空间投影, 从而有效利用先前的参数估计 $\hat{\theta}[i-1]$ 。另一方面, 如果 $\Omega[i]$ 具有满秩, 则零空间不存在, 倾斜零空间投影 di

总之, 公式 (27) 旨在通过使用从输入 $\Gamma[i]$ 到映射 $\Omega[i]$ 来获得动态参数的最优估计。在辨识过程中, 如果映射 $\Omega[i]$ 失去秩, 则通过倾斜零空间投影方案利用之前的参数估计, 最终提高辨识精度和收敛速度。下一节将通过若干仿真展示所提出辨识算法的有效性。

四、模拟

假设一个平面两连杆机械手正在如图所示在末端执行器处操作未知质量 M

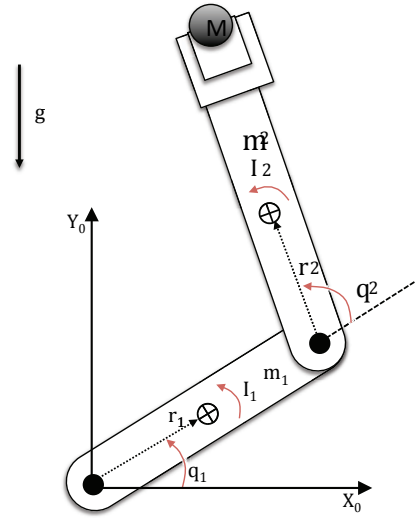


图 2: 典型平面两连杆机械手操作未知质量 M

在图2中, 为了验证所提出的识别算法。方程(1)的动态具有以下组成部分:

$$M(q) = \begin{bmatrix} I_1 + I_2 + m_2 l_1^2 + 2m_2 r_2 l_1 c_2 & I_2 + m_2 r_2 l_1 c_2 \\ I_2 + m_2 r_2 l_1 c_2 & I_2 \end{bmatrix} \quad (29)$$

$$C(q, \dot{q}) = \begin{bmatrix} -m_2 r_2 l_1 s_2 \dot{q}_2 & -m_2 r_2 l_1 s_2 (\dot{q}_1 + \dot{q}_2) \\ m_2 r_2 l_1 s_2 \dot{q}_1 & 0 \end{bmatrix} \quad (30)$$

$$g(q) = \begin{bmatrix} -m_1 r_1 g c_1 - m_2 l_1 g c_1 - m_2 r_2 g c_{12} \\ -m_2 r_2 g c_{12} \end{bmatrix} \quad (31)$$

$$\tau_f(q, \dot{q}) = \begin{bmatrix} F_{s1} \text{sign}(\dot{q}_1) + F_{c1} \dot{q}_1 \\ F_{s2} \text{sign}(\dot{q}_2) + F_{c2} \dot{q}_2 \end{bmatrix} \quad (32)$$

$$J(q) = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix} \quad (33)$$

$$f_{ext} = \begin{bmatrix} 0 \\ -Mg \end{bmatrix} \quad (34)$$

其中 I_1 和 I_2 分别是连杆 1 和 2 的转动惯量, m_1 和 m_2 分别是连杆 1 和 2 的质量, l_1 和 l_2 分别是连杆 1 和 2 的长度, r_1 和 r_2 分别是连杆 1 和 2 的质心位置, q_1 和 q_2 分别是连杆 1 和 2 的关节角度, F_{s1} 和 F_{s2} 分别是连杆 1 和 2 的静摩擦系数, F_{c1} 和 F_{c2} 分别是连杆 1 和 2 的库仑摩擦系数, g 表示重力加速度常数, $g = 9.8[m/s^2]$, 且

$$\begin{aligned} c_1 &= \cos(q_1) & c_2 &= \cos(q_2) \\ c_{12} &= \cos(q_1 + q_2) & s_1 &= \sin(q_1) \\ s_2 &= \sin(q_2) & s_{12} &= \sin(q_1 + q_2). \end{aligned}$$

对于仿真研究, 动态参数设置为 $l_1 = l_2 = 0.5[m]$ 、 $m_1 = m_2 = 1.0[kg]$ 、 $r_1 = r_2 = 0.2[m]$ 、 $I_1 = I_2 = 0.05[kg \cdot m^2]$ 、 $F_{s1} = F_{s2} = 0.1$ 、 $F_{c1} = F_{c2} = 0.1$ 和 $M = 0.1[kg]$ 。此外, 这些参数也被视为稍后要识别的不确定量。

A. Constructing Momentum Regressor

The parameter vector to be identified through the identification algorithm is defined as following form:

$$\theta \triangleq \begin{bmatrix} I_1 + m_2 l_1^2 \\ I_2 \\ m_1 r_1 + m_2 l_1 + M l_1 \\ m_2 r_2 + M l_2 \\ m_2 r_2 l_1 \\ F_{s1} \\ F_{s2} \\ F_{c1} \\ F_{c2} \end{bmatrix} \in \mathbb{R}^9, \quad (35)$$

where the exact result to be obtained will be

$$\theta^T = [0.3, 0.05, 0.75, 0.25, 0.1, 0.1, 0.1, 0.1, 0.1],$$

if the proposed identification algorithm is effective. As suggested in section II, we need to find two subregressors to construct the momentum regressor by using Eqs. (6) and (8) as follows:

$$W_1 = \begin{bmatrix} 0 & 0 & a_{13} & a_{14} & 0 & a_{16} & 0 & a_{18} & 0 \\ 0 & 0 & 0 & a_{24} & a_{25} & 0 & a_{27} & 0 & a_{29} \end{bmatrix} \quad (36)$$

where

$$\begin{aligned} a_{13} &= g c_1 & a_{14} &= g c_{12} \\ a_{16} &= -\text{sign}(\dot{q}_1) & a_{18} &= -\dot{q}_1 \\ a_{24} &= g c_{12} & a_{25} &= -s_2(\dot{q}_1 + \dot{q}_2)\dot{q}_1 \\ a_{27} &= -\text{sign}(\dot{q}_2) & a_{29} &= -\dot{q}_2 \end{aligned}$$

and

$$W_2 = \begin{bmatrix} b_{11} & b_{12} & 0 & 0 & b_{15} & 0 & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 & b_{25} & 0 & 0 & 0 & 0 \end{bmatrix} \quad (37)$$

where

$$\begin{aligned} b_{11} &= \dot{q}_1 & b_{12} &= \dot{q}_1 + \dot{q}_2 \\ b_{15} &= c_2(2\dot{q}_1 + \dot{q}_2) & b_{22} &= \dot{q}_1 + \dot{q}_2 \\ b_{25} &= c_2\dot{q}_1. \end{aligned}$$

Now we can get the momentum regressor as either continuous-time form of Eq. (11) or discrete-time form of Eq. (13). To simplify the simulation and the related discussions, the control input torques are set as follows:

$$\tau[i] = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{for } i = 1, 2, \dots$$

For the simulation study, the Eqs. (13) and (14) will be utilized in the following subsection.

B. Identification Result of the Proposed Scheme

The pseudo code of the proposed identification scheme in the section III is summarized as follows:

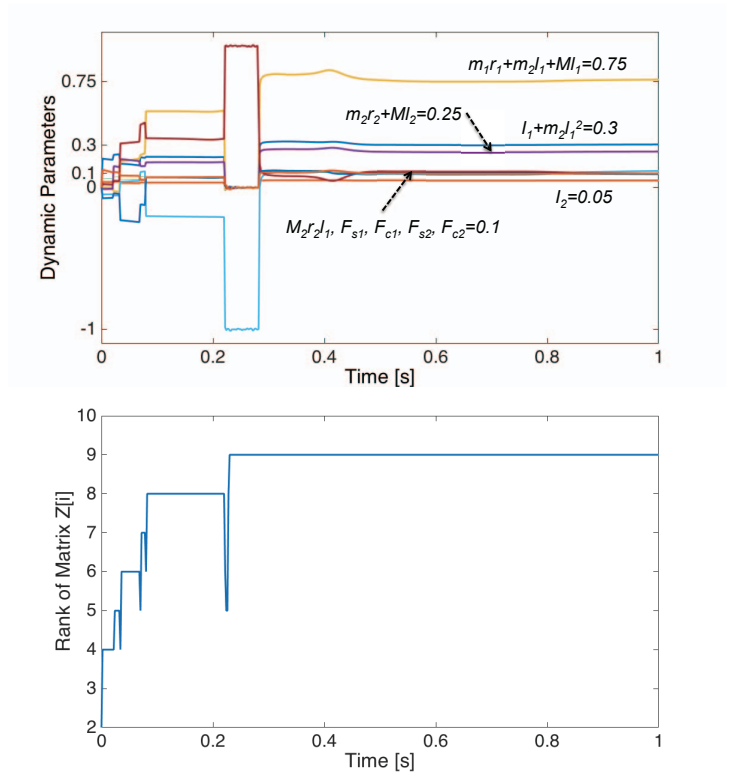


Fig. 3: Simulation result obtained by using the proposed scheme when the initial $\Omega[0] = 0 \cdot I$

```

 $\Omega[0] = 0 \cdot I$       (note that  $\Omega[i] \in \mathbb{R}^{9 \times 9}$ )
 $\Gamma[0] = 0$         (note that  $\Gamma[i] \in \mathbb{R}^9$ )
 $\delta t = 2[ms]$ 
 $n = \frac{t_f}{\delta t}$       ( $t_f$  is a terminal time of simulation)
for  $i = 1 : n$ 
    time  $t = i \cdot \delta t$ 
    generate  $\tau[t] = 1$       (note that  $\tau[i] \in \mathbb{R}^2$ )
    solve Eq. (1):  $q[t], \dot{q}[t]$ 
    construct momentum regressor Eq. (12):
         $Y_H[i], u[i]$  (note that  $Y_H[i] \in \mathbb{R}^{2 \times 9}$  and  $u[i] \in \mathbb{R}^2$ )
         $\Omega[i] = \Omega[i-1] + Y_H^T[i] Y_H[i]$ 
         $\Gamma[i] = \Gamma[i-1] + Y_H^T[i] u[i]$ 
         $W[i] = (I + \Omega^T[i] \Omega[i])^{-1}$  (note that  $W[i] \in \mathbb{R}^{9 \times 9}$ )
         $Z[i] = (\Omega[i] W[i] \Omega^T[i])^+$  (note that  $Z[i] \in \mathbb{R}^{9 \times 9}$ )
         $\Omega_W^+[i] = W[i] \Omega[i]^T Z[i]$ 
         $\hat{\theta}[i] = \Omega_W^+[i] \Gamma[i] + (I - \Omega_W^+[i] \Omega[i]) W[i] \hat{\theta}[i-1]$ 
end

```

To begin with, the identification result obtained by using the proposed scheme is shown in Fig. 3(a) when the zero initial condition, namely $\Omega[0] = 0 \cdot I$, was used. Here we can confirm that the proposed identification scheme has

A. Constructing Momentum Regressor

通过识别算法要识别的参数向量定义如下形式：

$$\theta \triangleq \begin{bmatrix} I_1 + m_2 l_1^2 \\ I_2 \\ m_1 r_1 + m_2 l_1 + M l_1 \\ m_2 r_2 + M l_2 \\ m_2 r_2 l_1 \\ F_{s1} \\ F_{s2} \\ F_{c1} \\ F_{c2} \end{bmatrix} \in \mathbb{R}^9, \quad (35)$$

将要得到确切结果的地方

$$\theta^T = [0.3, 0.05, 0.75, 0.25, 0.1, 0.1, 0.1, 0.1, 0.1],$$

如果所提出的识别算法是有效的。如第二节所建议的，我们需要找到两个子回归器，通过使用公式 (6) 和 (8) 来构建动量回归器，如下所示：

$$W_1 = \begin{bmatrix} 0 & 0 & a_{13} & a_{14} & 0 & a_{16} & 0 & a_{18} & 0 \\ 0 & 0 & 0 & a_{24} & a_{25} & 0 & a_{27} & 0 & a_{29} \end{bmatrix} \quad (36)$$

哪里

$$\begin{aligned} a_{13} &= g c_1 & a_{14} &= g c_{12} \\ a_{16} &= -\text{sign}(\dot{q}_1) & a_{18} &= -\dot{q}_1 \\ a_{24} &= g c_{12} & a_{25} &= -s_2(\dot{q}_1 + \dot{q}_2)\dot{q}_1 \\ a_{27} &= -\text{sign}(\dot{q}_2) & a_{29} &= -\dot{q}_2 \end{aligned}$$

和

$$W_2 = \begin{bmatrix} b_{11} & b_{12} & 0 & 0 & b_{15} & 0 & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 & b_{25} & 0 & 0 & 0 & 0 \end{bmatrix} \quad (37)$$

哪里

$$\begin{aligned} b_{11} &= \dot{q}_1 & b_{12} &= \dot{q}_1 + \dot{q}_2 \\ b_{15} &= c_2(2\dot{q}_1 + \dot{q}_2) & b_{22} &= \dot{q}_1 + \dot{q}_2 \\ b_{25} &= c_2\dot{q}_1. \end{aligned}$$

现在我们可以得到动量回归器，它可以是方程(11)的连续时间形式，也可以是方程(13)的离散时间形式。为了简化仿真和相关讨论，控制输入力矩设置如下：

$$\tau[i] = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{for } i = 1, 2, \dots$$

在模拟研究中，方程 (13) 和 (14) 将在以下小节中使用。

B. Identification Result of the Proposed Scheme

第三节中提出的识别方案的伪代码总结如下：

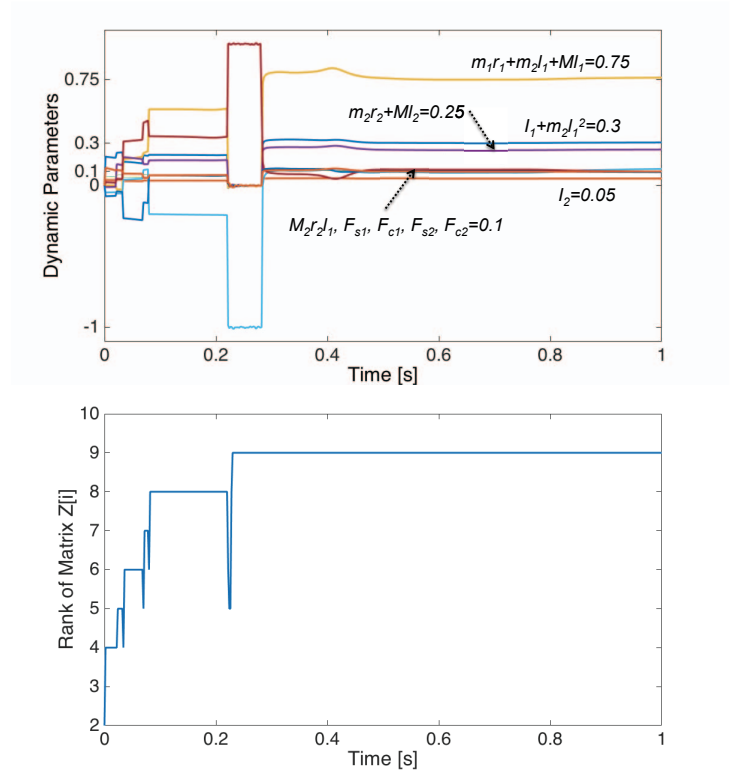


图3: 使用所提出方案在初始 $\Omega[0] = 0 \cdot I$ 时获得的仿真结果

```

 $\Omega[0] = 0 \cdot I$  (note that  $\Omega[i] \in \mathbb{R}^{9 \times 9}$ )
 $\Gamma[0] = 0$  (note that  $\Gamma[i] \in \mathbb{R}^9$ )
 $\delta t = 2[ms]$ 
 $n = \frac{t_f}{\delta t}$  ( $t_f$  is a terminal time of simulation)
for  $i = 1 : n$ 
    time  $t = i \cdot \delta t$ 
    generate  $\tau[t] = 1$  (note that  $\tau[i] \in \mathbb{R}^2$ )
    solve Eq. (1):  $q[t], \dot{q}[t]$ 
    construct momentum regressor Eq. (12):
         $Y_H[i], u[i]$  (note that  $Y_H[i] \in \mathbb{R}^{2 \times 9}$  and  $u[i] \in \mathbb{R}^2$ )
         $\Omega[i] = \Omega[i-1] + Y_H^T[i] Y_H[i]$ 
         $\Gamma[i] = \Gamma[i-1] + Y_H^T[i] u[i]$ 
         $W[i] = (I + \Omega^T[i] \Omega[i])^{-1}$  (note that  $W[i] \in \mathbb{R}^{9 \times 9}$ )
         $Z[i] = (\Omega[i] W[i] \Omega^T[i])^+$  (note that  $Z[i] \in \mathbb{R}^{9 \times 9}$ )
         $\Omega_W^+[i] = W[i] \Omega[i]^T Z[i]$ 
         $\hat{\theta}[i] = \Omega_W^+[i] \Gamma[i] + (I - \Omega_W^+[i] \Omega[i]) W[i] \hat{\theta}[i-1]$ 
end

```

首先，使用所提出的方案在使用零初始条件，即 $\Omega[0] = 0 \cdot I$ 时，得到的识别结果如图 3(a) 所示。在这里我们可以确认所提出的识别方案具有

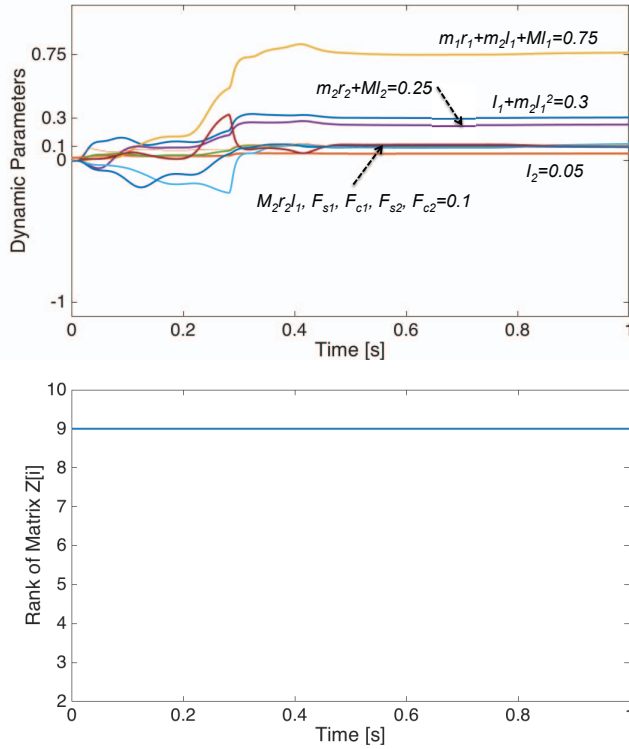


Fig. 4: Simulation result obtained by using the proposed scheme when the initial $\Omega[0] = 0.0001 \cdot I$

produced the exact result after undergoing several discontinuous phenomena. To illustrate the reason of discontinuities, we have suggested the rank variation graph of the matrix $Z[i]$ as shown in Fig. 3(b). Whenever the rank of matrix $Z[i]$ varies, the identification results jump to another values, ultimately until arriving at the exact results. After being satisfied with the full rank condition (here 9) of the matrix $Z[i]$, the identification results are fastly stabilized towards to the exact values.

On the other hand if the initial conditions are assigned to arbitrary small values, such as $\Omega[0] = 0.0001 \cdot I$, then the discontinuity phenomena disappear as shown in Fig. 4. Furthermore, if we increase the initial condition of $\Omega[0]$ to $0.01 \cdot I$, then more stable and smooth results are obtained as shown in Fig. 5 but the convergence speed becomes also slower compared with the identification results obtained from the zero initial condition. These good identification results could be obtained thanks to the momentum regressor and the identification scheme proposed newly in the paper.

V. CONCLUSION

The paper has proposed the momentum regressor as new form and the identification algorithm based on the Lagrange multiplier and optimization. The momentum regressor was suggested through the integration of the conventional Lagrange regressor. It has a robustness property against the measurement noise because it made use of the numerical integration. Finally the simulations were conducted to show the effectiveness of the proposed method.

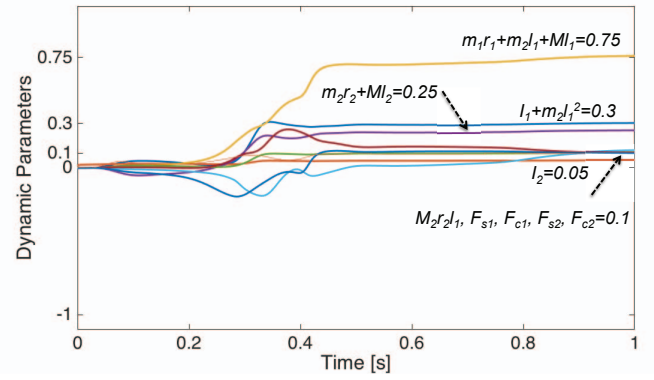


Fig. 5: Simulation result obtained by using the proposed scheme when the initial $\Omega[0] = 0.01 \cdot I$

For future works, we are to develop an adaptive control based on the proposed identification scheme. In addition, the comparative studies with existing adaptive control schemes are implemented to show the relative superiority of the proposed scheme.

REFERENCES

- [1] T. Söderström and P. Stoica, *System Identification*, Prentice Hall, 1989.
- [2] P. K. Khosla, Categorization of Parameters in the Dynamics Robot Model, *IEEE Trans. on Robotics and Automation*, vol. 5, No. 3, pp. 261-268, 1989.
- [3] C. C. de Wit, B. Sciliano, and G. Bastin, *Theory of Robot Control*, Springer, 1996.
- [4] M. W. Spong, S. Hutchinson and M. Vidyasagar, *Robot Modeling and Control*, Wiley, 2005.
- [5] F. L. Lewis, C. T. Abdallah and D. M. Dawson, *Control of Robot Manipulators*, Macmillan, 1993.
- [6] R. Featherstone, *Rigid Body Dynamics Algorithms*, Springer, 2008.
- [7] J. Naudet, and D. Lefeber, General Formulation of an Efficient Recursive Algorithm Based on Canonical Momenta for Forward Dynamics of Closed-Loop Multibody Systems, *ASME 2005 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, pp. 177-185, 2005.
- [8] S. J. Leon, *Linear Algebra with Application*, 8th Edition, Pearson, 2010.
- [9] J. R. Magnus and H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, John Wiley & Sons, 1998.
- [10] Y. Choi and J. Cheong, "New Expressions of 2 x 2 Block Matrix Inversion," *IEEE Transactions on Automatic Control*, vol. 54, No. 11, pp. 2648-2653, 2009.
- [11] B. S. Chen, Y. C. Chang and T. C. Lee, "Adaptive Control in Robotic Systems with $\mathcal{H}_\infty$ Tracking Performance", *Automatica*, vol. 33, No. 2, pp. 227-234, 1997.
- [12] J.-J. E. Slotine and W. Li, Composite Adaptive Control of Robot Manipulators, *Automatica*, vol. 25, No. 4, pp. 509-519, 1989.
- [13] R. M. Rangayyan, *Biomedical Signal Analysis*, IEEE Press, 2002.

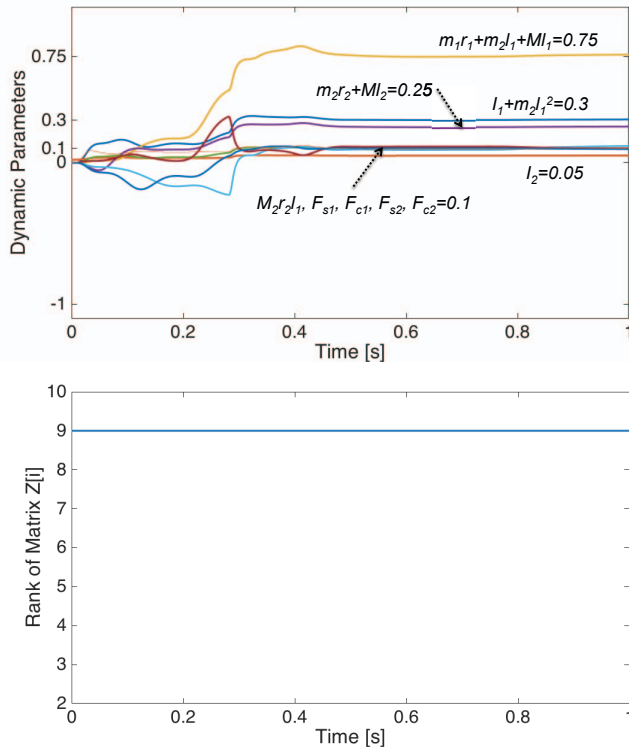


图4: 当初始 $\Omega[0] = 0.0001 \cdot I$ 时, 使用所提出方案获得的仿真结果

在经历了几次不连续现象后, 产生了精确的结果。为了说明不连续的原因, 我们提出了矩阵 $Z[i]$ 的秩变化图, 如图 3(b) 所示。每当矩阵 $Z[i]$ 的秩发生变化时, 识别结果会跳跃到另一个值, 最终达到精确结果。在满足矩阵 $Z[i]$ 的满秩条件 (此处为 9) 后, 识别结果会迅速稳定到精确值。

另一方面, 如果初始条件被赋值为任意小的数值, 例如 $\Omega[0] = 0.0001 \cdot I$, 那么如图 4 所示, 不连续现象将消失。此外, 如果我们将 $\Omega[0]$ 的初始条件增加到 $0.01 \cdot I$, 则如图 5 所示, 可以得到更加稳定和平滑的结果, 但与从零初始条件获得的识别结果相比, 收敛速度也会变慢。这些良好的识别结果得益于本文新提出的动量回归器和识别方案。

五、结论

本文提出了动量回归器作为一种新形式, 以及基于拉格朗日乘子和优化的识别算法。通过对传统拉格朗日回归器的积分, 提出了动量回归器。由于使用了数值积分, 它对测量噪声具有鲁棒性。最后进行了仿真实验, 以展示所提出方法的有效性。

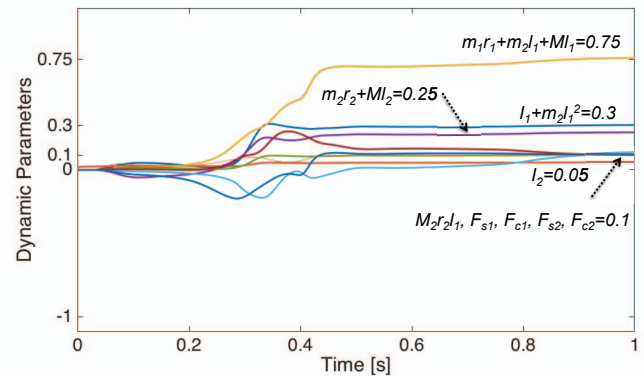


图5: 使用所提出方案在初始 $\Omega[0] = 0.01 \cdot I$ 时获得的仿真结果

在未来的工作中, 我们将基于所提出的识别方案开发一种自适应控制。此外, 已经实施了与现有自适应控制方案的比较研究, 以显示所提出方案的相对优越性。

参考文献

- [1] T. Söderström 和 P. Stoica, *System Identification*, Prentice Hall, 1989.
- [2] P. K. Khosla, 《机器人动力学模型参数的分类》, *IEEE Trans. on Robotics and Automation*, 第5卷, 第3期, 页 261-268, 1989.
- [3] C. C. de Wit, B. Sciliano, 和 G. Bastin, *Theory of Robot Control*, Springer, 1996.
- [4] M. W. Spong, S. Hutchinson 和 M. Vidyasagar, *Robot Modeling and Control*, Wiley, 2005.
- [5] F. L. Lewis, C. T. Abdallah 和 D. M. Dawson, *Control of Robot Manipulators*, Macmillan, 1993.
- [6] R. Featherstone, *Rigid Body Dynamics Algorithms*, Springer, 2008.
- [7] J. Naud et, 和 D. Lefeber, 《基于标准动量的闭环多体系统前向动力学高效递归算法的一般公式》, *ASME 2005 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, 页 177-185, 2005.
- [8] S. J. Leon, *Linear Algebra with Application, 8th Edition*, Pearson, 2010.
- [9] J. R. Magnus 和 H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, John Wiley & Sons, 1998.
- [10] Y. Choi 和 J. Cheong, “2 x 2 块矩阵求逆的新表达式,” *IEEE Transactions on Automatic Control*, 第54卷, 第11期, 页 2648-2653, 2009.
- [11] B. S. Chen, Y. C. Chang 和 T. C. Lee, “带有 $\mathcal{H}_\infty$ 跟踪的机器人系统自适应控制”